

Lecture -14

Filters:-

N/w + to components present in the N/w filters are classified as

(I) Active Filter

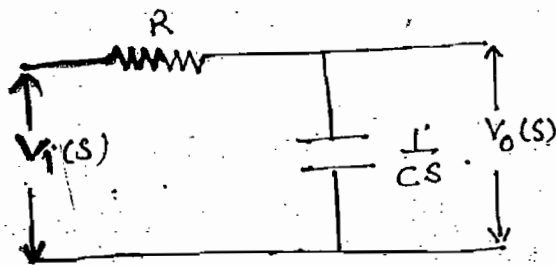
(II) Passive Filter

- Active filter consist of op-amp and capacitor
- In the active filter it is possible to inc. the gain of the system
- Generally inductor is not preferred in the active filter since its size is bulky and cost is high
- Passive filter consist of LC series and parallel sections (Reactive N/w)
- Based on the operating frequency filters are classified as

- (I) LPF (II) HPF (III) BPF (IV) BEF (Band stop)
(V) All pass

LPF:-

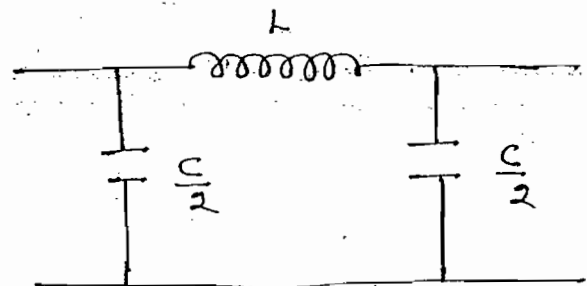
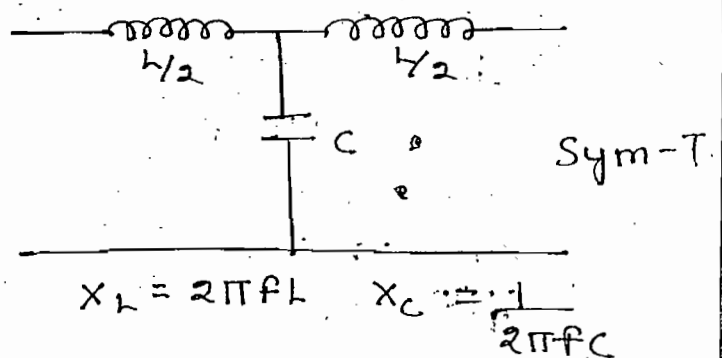
First order



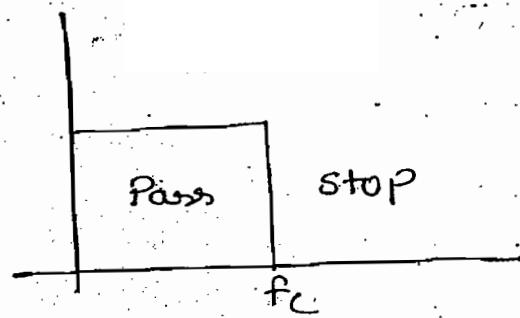
$$V_o(s) = V_i(s) \frac{\frac{1}{Cs}}{R + \frac{1}{Cs}}$$

$$\Rightarrow \boxed{\frac{V_o(s)}{V_i(s)} = \frac{1}{1 + RCs}}$$

Second order



Sym-Π



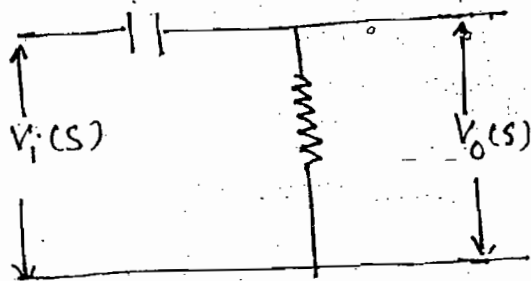
→ Inductor provide low resistance path and capacitor bypass higher frequency

$L \rightarrow$

HPF :-

→ For lower freq. capacitor provides high impedance

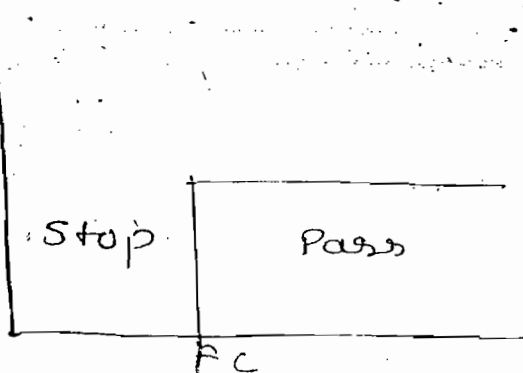
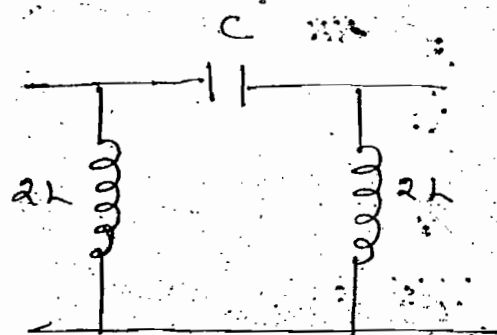
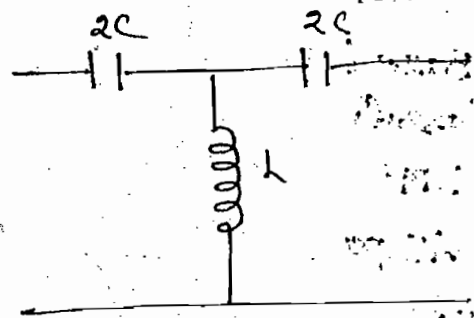
First Order



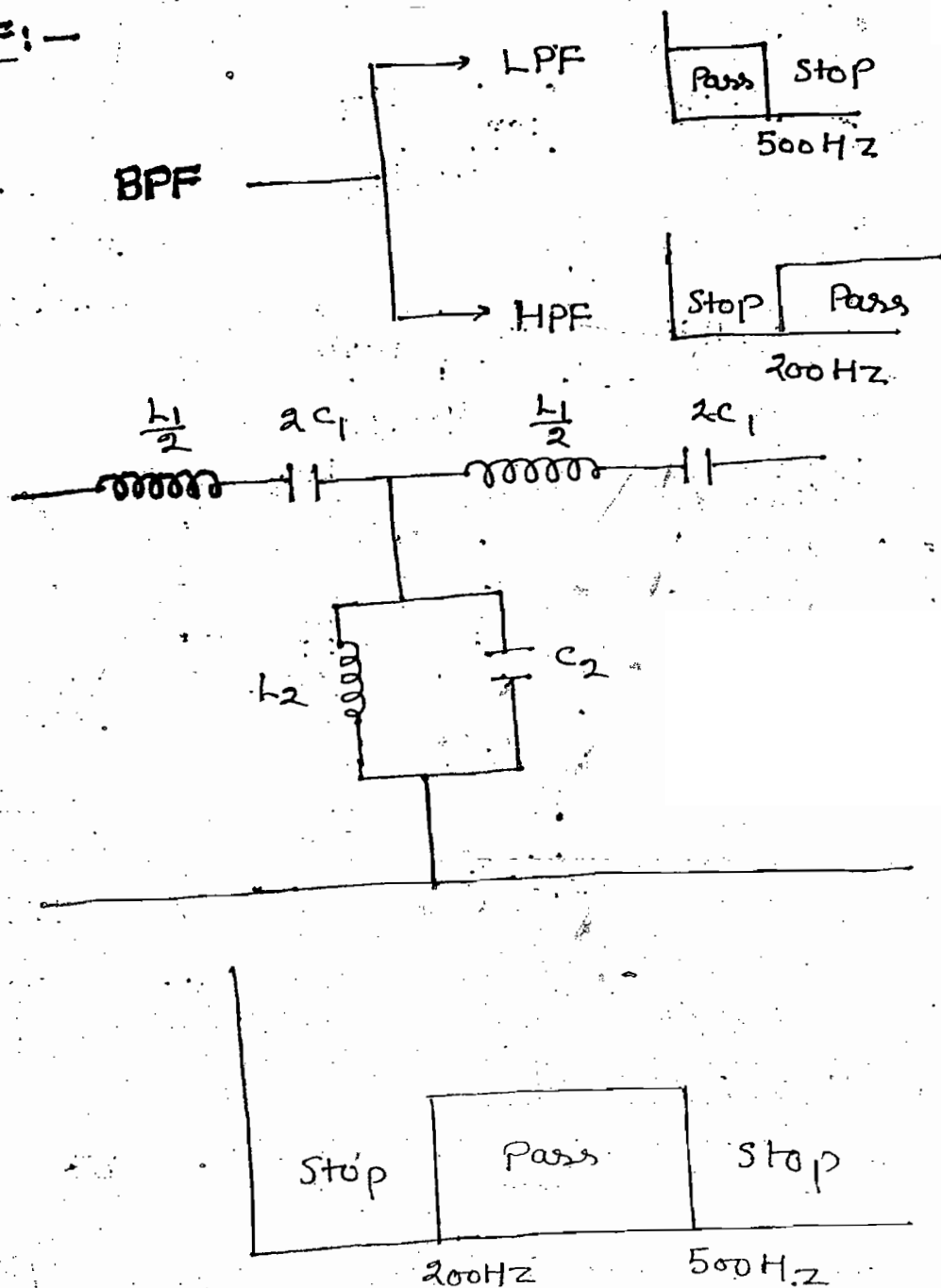
$$V_o(s) = V_i(s) \frac{R}{R + \frac{1}{Cs}}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{Rcs}{Rcs + 1}$$

Second Order

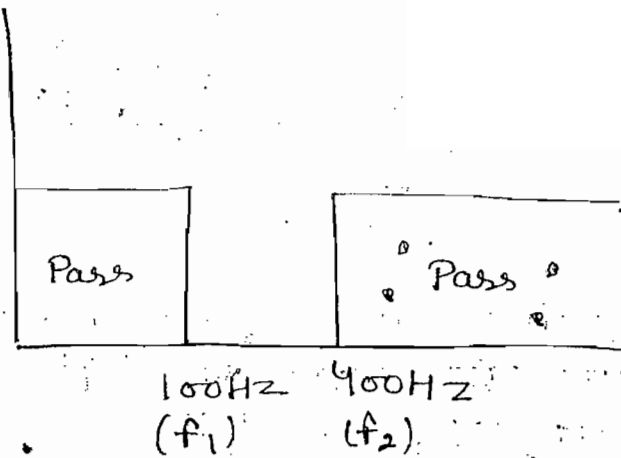
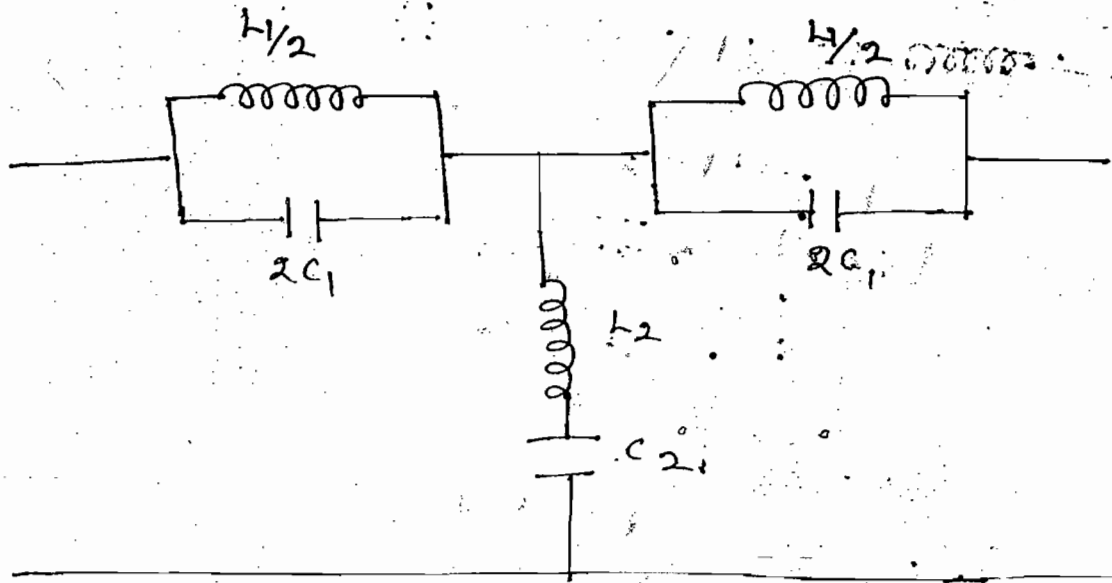
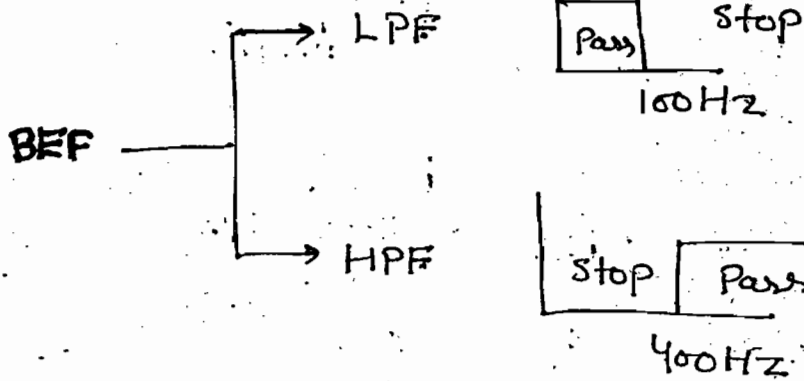


BPF:-



- BPF is obtained with a combination of LPF & HPF and cut-off freq. of LPF should be greater than cut-off freq. of HPF.
- In the BPF first cut-off frequency is calculated w.r.t series resonance and second cut-off freq. is calculated w.r.t parallel resonance.

BEP :-



→ BEP is obtained with a combination of LPF and HPF and cut-off frequency of high pass filter should be greater than cut-off freq. of low pass filter

Transfer Function of first order filter:-

$\frac{1}{1+Ts}$	→ LPF
$\frac{Ts}{1+Ts}$	→ HPF
$\frac{1-Ts}{1+Ts}$	→ All pass

Note!— In all pass filter zero's are present in the right half of plane and poles are present in left half of plane

Transfer Function of second order filter:-

$$\frac{P}{s^2 + as + b} \rightarrow \text{LPF}$$

$$\frac{Ps^2 + q}{s^2 + as + b} \rightarrow \text{BEF}$$

$$\frac{Ps^2}{s^2 + as + b} \rightarrow \text{HPF}$$

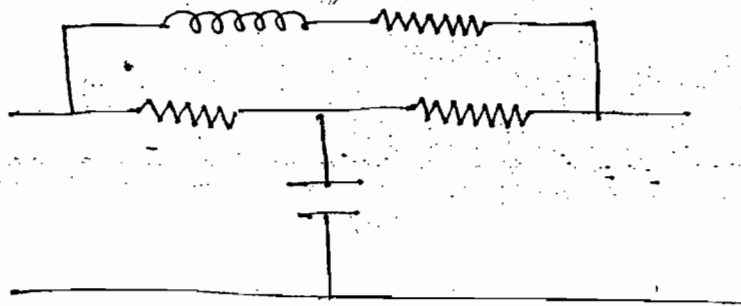
$$\frac{s^2 - Ps + q}{s^2 + as + b} \rightarrow \text{All Pass}$$

$$\frac{Ps}{s^2 + as + b} \rightarrow \text{BPF}$$

Note!—

Poles and zeroes are symmetrical about $j\omega$ -axis (All pass)

Ques! Identify type of the filter of the ckt shown



Soln:-

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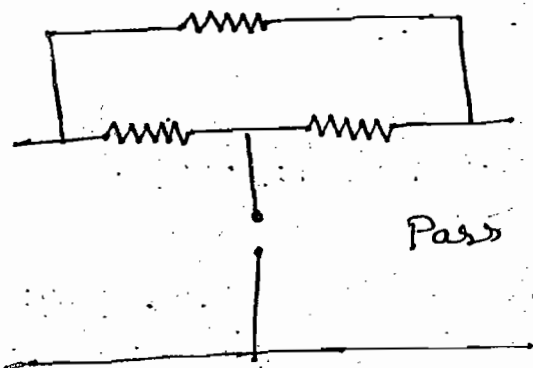
$$f = 0$$

$$X_L = 2\pi fL = 0$$

$$L \rightarrow SC$$

$$X_C = \frac{1}{2\pi fC} = \infty$$

$$C \rightarrow OC$$



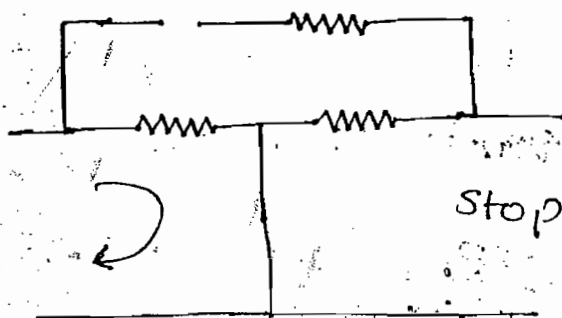
$$f = \infty$$

$$X_L = 2\pi fL = \infty$$

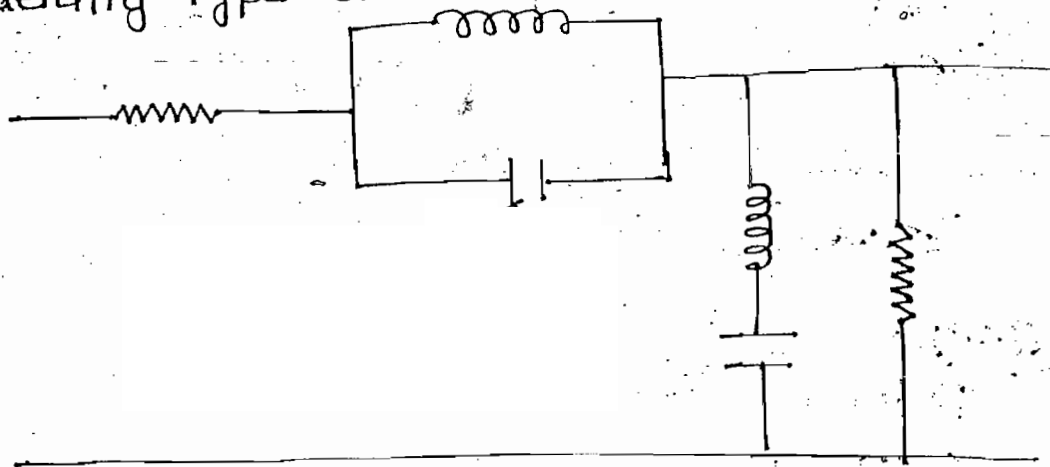
$$L \rightarrow OC$$

$$X_C = \frac{1}{2\pi fC} = 0$$

$$C \rightarrow SC$$



Ques:- Identify type of the filter of ckt. shown.



Soln:-

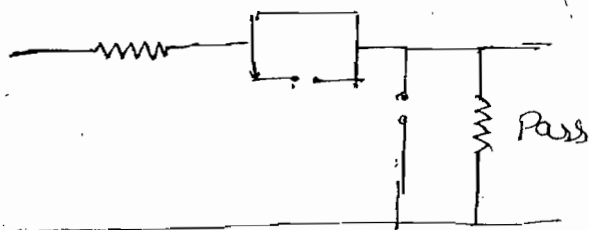
$$f = 0$$

$$X_L = 2\pi fL = 0$$

$$L \rightarrow SC$$

$$X_C = \frac{1}{2\pi fC} = \infty$$

$$C \rightarrow OC$$



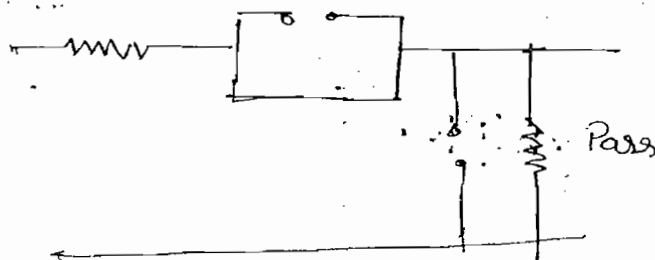
$$f = \infty$$

$$X_L = 2\pi fL = \infty$$

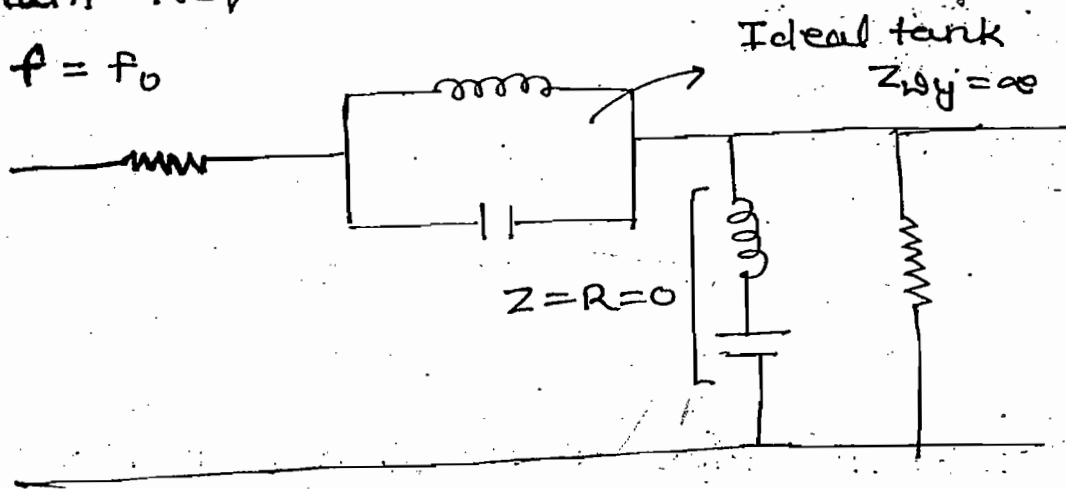
$$L \rightarrow OC$$

$$X_C = \frac{1}{2\pi fC} = 0$$

$$C \rightarrow SC$$



For BEF & All pass filter draw eq. ckt at resonant freq.

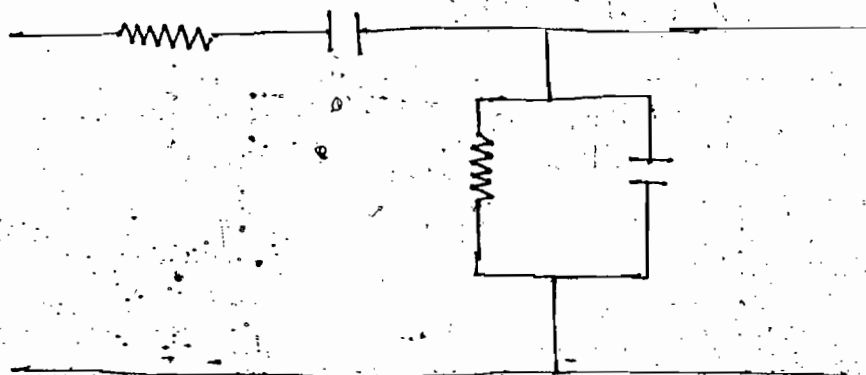


→ BEF , Ans

Note:-

When BEF eliminates only few frequency then it is also called as Notch filter.

ques:- Identify type of the filter of the ckt shown

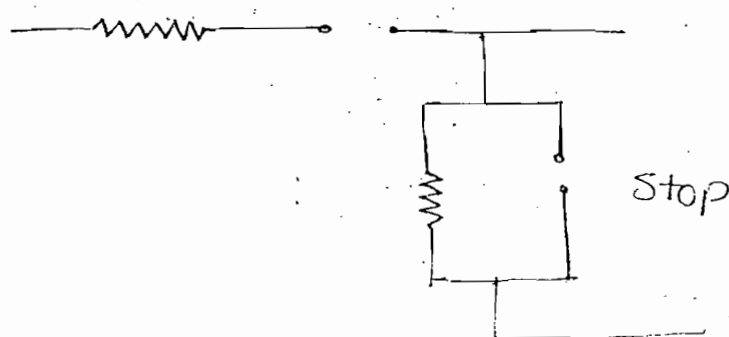


Soln:- $f = 0$

$$X_L = 2\pi fL = 0$$

$h \rightarrow S.C$

$$X_C = \frac{1}{2\pi fC} = \infty$$



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$C \rightarrow O.C$

$$f = \infty$$

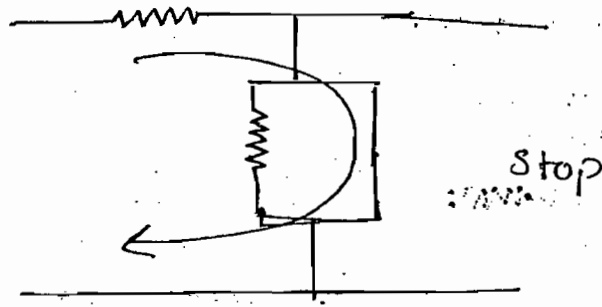
$$X_L = 2\pi fL = \infty$$

$$L \rightarrow \text{O.C}$$

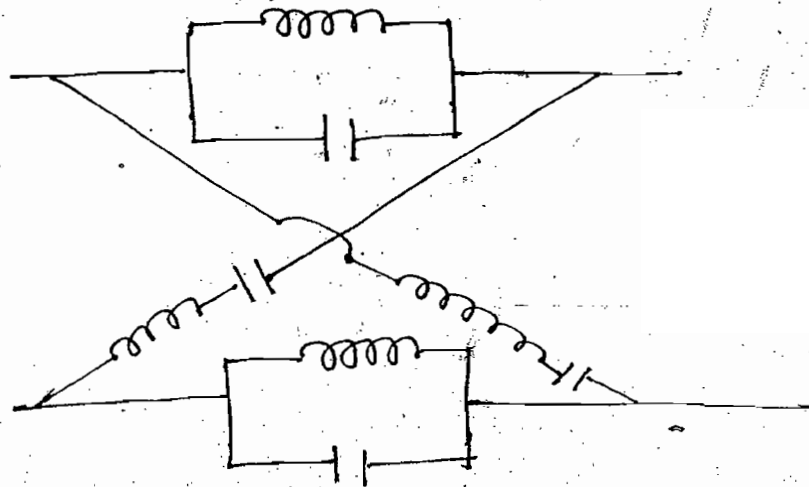
$$X_C = \frac{1}{2\pi fC} = 0$$

$$C \rightarrow \text{S.C}$$

→ BPF



Ques:- Identify type of the filter of the ckt shown



Ans:- All pass ($f = 0, f = \infty, f = f_0$)